

C++ OOP Project: Quest Archive Manager

Medium-Difficulty Project Specification

Project Goal

Build a console-based application for managing a guild archive of adventurers and quests. The project should use Object-Oriented Programming and be organized into separate files.

Required File Structure

- Adventurer.h - class declaration
- Adventurer.cpp - class implementation
- main.cpp - menu system and application logic
- quests.txt - persistent data storage

Class Design

Create an Adventurer class with private data members such as name, level, questsCompleted, goldEarned, and rank.

Suggested Methods

- 1 Constructors
- 2 recordQuest()
- 3 display()
- 4 updateRank()
- 5 getName()
- 6 getGoldEarned()
- 7 getLevel()
- 8 saveToFile() style helper methods if desired

Required Programming Concepts

- 1 Classes and objects
- 2 Header and implementation files
- 3 Functions
- 4 Vectors or arrays
- 5 Loops
- 6 Switch statements
- 7 Conditionals
- 8 Searching
- 9 Sorting
- 10 File I/O
- 11 Input validation

Program Features

- 1 Add Adventurer
- 2 Record Quest
- 3 Display All Adventurers
- 4 Search by Name
- 5 Sort by Gold Earned
- 6 Generate Statistics
- 7 Delete Adventurer
- 8 Save Data
- 9 Load Data
- 10 Exit Program

quests.txt Format

Store one adventurer per line using CSV-style formatting:

Aria,12,25,5400,Elite

Kael,7,11,1800,Veteran

Statistics Requirements

- 1 Total adventurers
- 2 Highest gold earned
- 3 Average level
- 4 Total quests completed
- 5 Top-ranked adventurer

Stretch Goals

- 1 Automatic rank progression
- 2 Achievement badges
- 3 Quest recommendation system
- 4 Multiple save files
- 5 Additional sorting options

Success Criteria

The application should compile cleanly, preserve data using `quests.txt`, use OOP correctly, separate interface and implementation files, and provide a robust menu-driven experience.